GTC4Lusso USA - Suspension damping control system - Latest Update: 2021-01-26

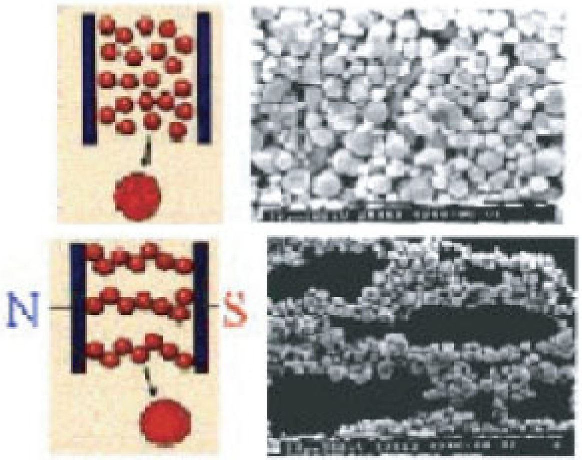
D4.07 Suspension damping control system

System function

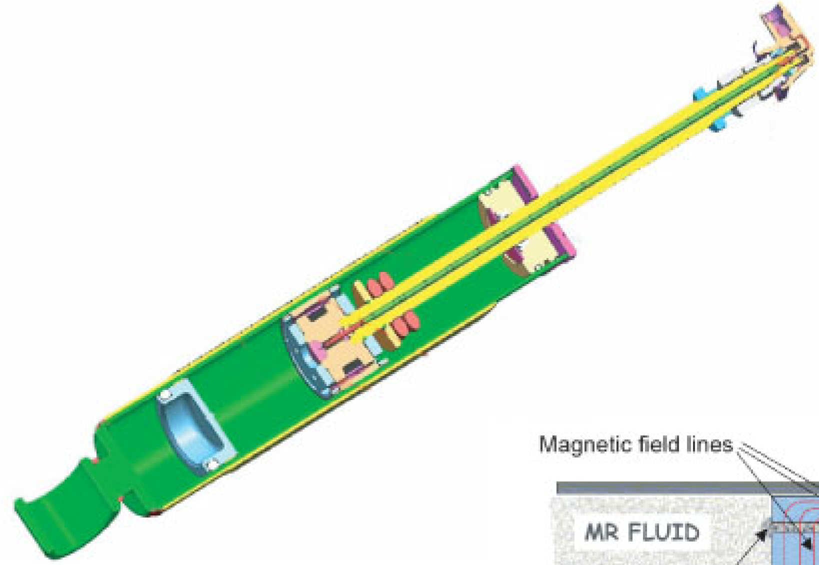
This vehicle is equipped with the latest generation MagneRide™ magnetorheologically controlled suspension, a system that continuously and automatically controls suspension damping developed by Delphi and honed by Ferrari.

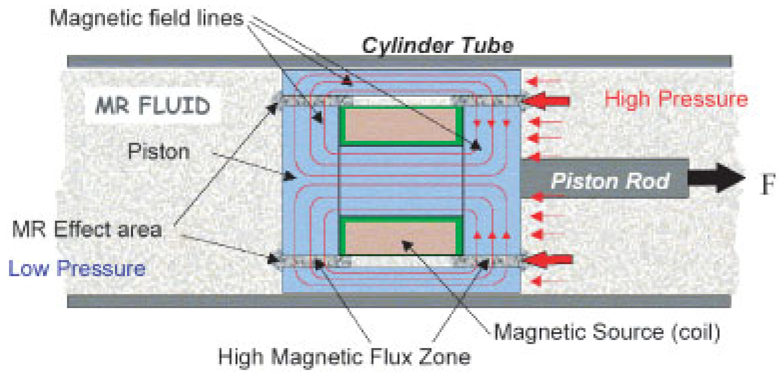
By processing data received from the vehicle dynamics sensors measuring the movements of the bodyshell, the ECU determines the road surface conditions and instantaneously adapts the response of the suspension accordingly, by varying the control current for each shock absorber.

These sensors allow the ECU to calculate vehicle speed, vertical and lateral acceleration, steering angle and instantaneous brake system pressure, and therefore control suspension damping in relation to these parameters.



Damping at each wheel is adjusted by a magnetorheological (MR) fluid which changes in density in relation to a magnetic field applied to the shock absorber.





The magnetorheological (MR) fluid is an oil containing a suspension of spherical ferromagnetic particles measuring a few microns in diameter.

Using an electric current to apply a magnetic field to the oil, the viscosity of the oil may be varied.

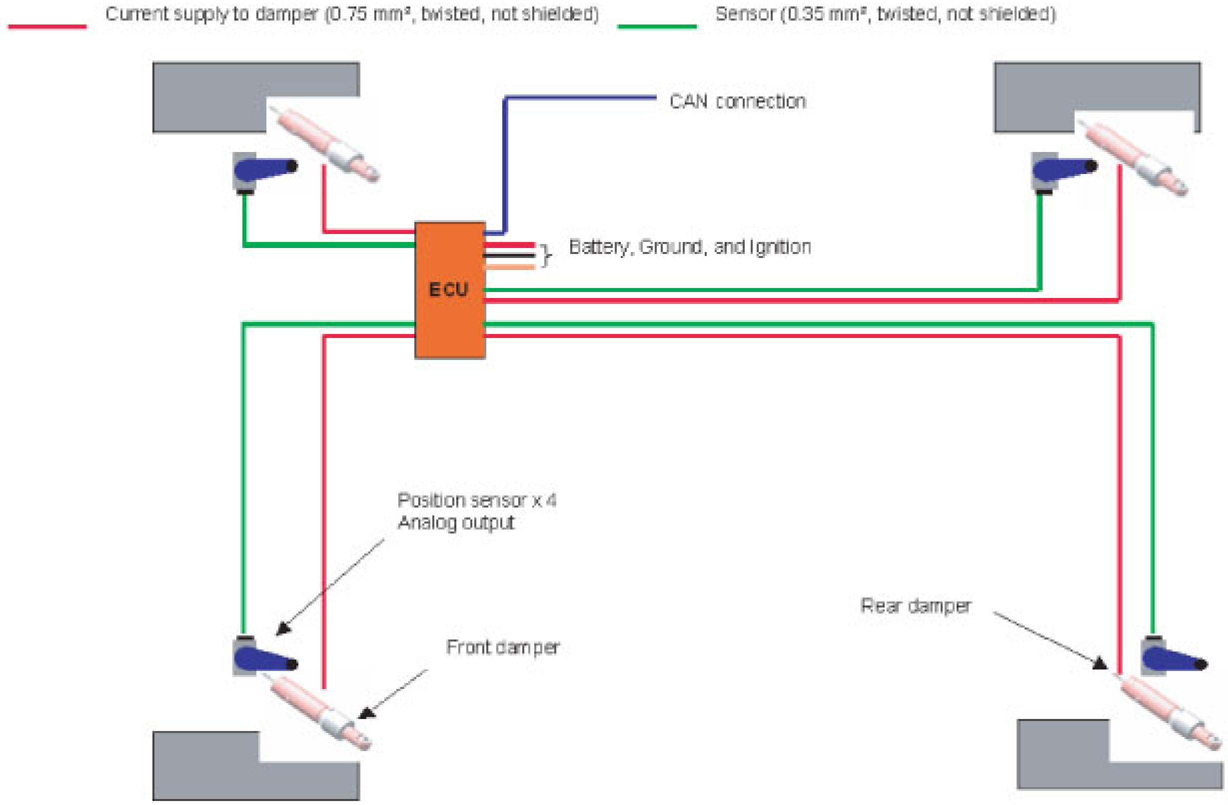
By varying the magnetic field applied to the MR fluid flowing through a duct, its pressure and flow rate characteristics may be adjusted in real time without the use of any external mechanical moving parts or actuators.

As well as ensuring the perfect balance between handling and comfort at all times, through specific calibrations selectable from the Manettino driving mode selector, this system can also shift the bias slightly towards one or the other of these two aspects.

Three different calibration levels are used on this vehicle:

- Level 1 (ICE): calibration optimised for maximum traction in low and very low grip conditions, such as snow or ice (ICE Manettino setting).
- Level 2 (COMFORT): a slightly stiffer calibration optimised to absorb bumps better and to offer improved grip in wet conditions (WET and COMFORT Manettino settings).
- Level 3 (SPORT): an even stiffer calibration optimised for sports and high speed use (in medium to high grip conditions or on the track) without drastically compromising comfort (SPORT and ESC OFF Manettino settings).

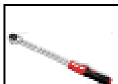
System layout



The system consists of:

- **4 NRRPS** (Non-Contacting Ratiometer Rotational Position Sensor) sensors installed on the suspension of each wheel, in which the fixed part of the potentiometer is mounted on the chassis (sprung mass) while the rotating part of the potentiometer moves together with the suspension. The sensors transmit a linear signal to the NCS ECU proportional to the angle of movement.
- **NCS** ECU, installed under the hatch cover in the passenger side footrest mat which, in relation to the signals received from the four NRRPS sensors, the data provided by the ECUs connected to the NCS and the Manettino setting, controls the damping function of the shock absorbers.
- **4 magnetorheological shock absorbers**, with one for each wheel suspension, controlled by the NCS ECU in relation to the signals received by the latter.

Replacing the suspension movement sensors

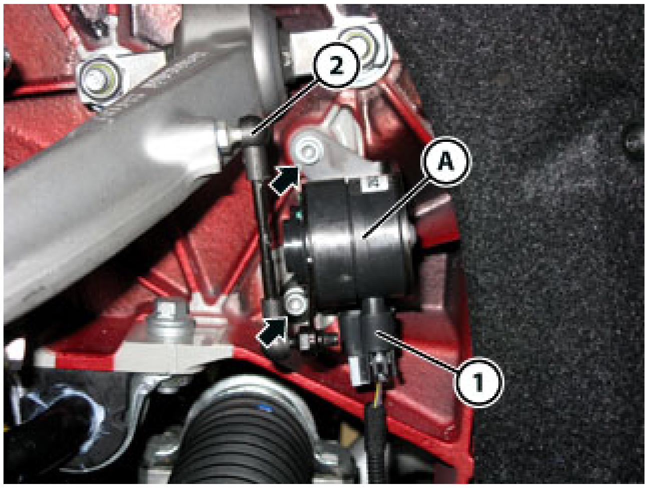
			
Tightening torque		**Nm**	**Class**
Fastening the NRRPS movement sensors	Screw / Nut	5 Nm	B

➢ Disconnect the battery (F2.01).

Front sensors

- Remove the complete wheels (D2.01).

Remove the front wheels.



- Disconnect the connector **(1)**.
- Detach the tie-rod **(2)**.
- Undo the screws indicated.
- Remove the sensor **(A)**, complete with the respective mounting bracket, and replace.
- Fit the new sensor **(A)**, complete with the respective mounting bracket.
- Tighten the screws indicated.

Tightening torque	Nm	Class
Screw / Nut	5 Nm	B

- Attach the tie-rod **(2)**.
- Connect the connector **(1)**.

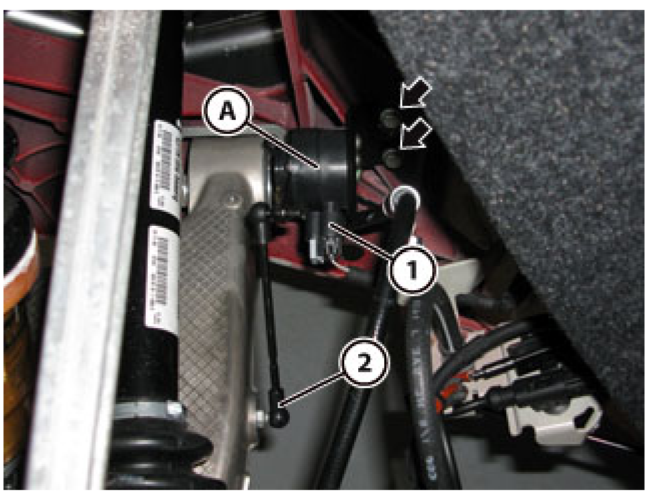
- Refit the complete wheels (D2.01).

Refit the front wheels.

Rear sensors

- Remove the complete wheels (D2.01).

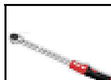
Remove the rear wheels.



- Disconnect the connector **(1)**.
- Detach the tie-rod **(2)**.
- Undo the indicated screws/nuts.

FROM Assembly No. 111223, the fastener screws have been replaced by nuts

- Remove the sensor **(A)**, complete with the respective mounting bracket, and replace.
- Fit the new sensor **(A)**, complete with the respective mounting bracket.
- Tighten the indicated screws/nuts.



Tightening torque	Nm	Class
Screw / Nut	5 Nm	B

- Attach the tie-rod **(2)**.
- Connect the connector **(1)**.

- Refit the complete wheels (D2.01).

Refit the rear wheels.

- Connect the battery (F2.01).

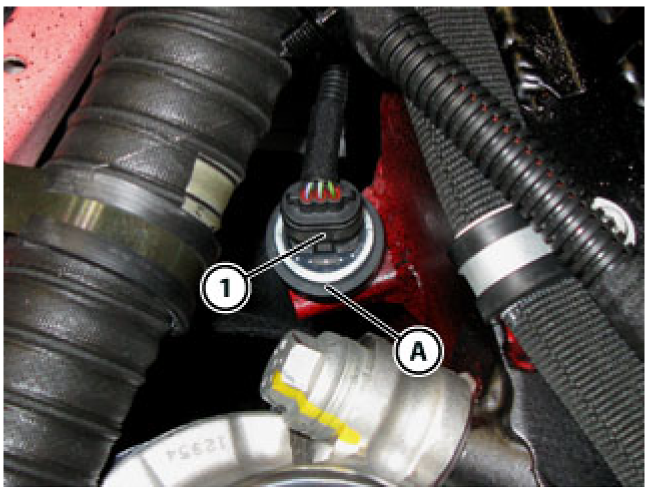
Replacing the Magneride accelerometers

Tightening torque		**Nm**	**Class**
Fastening Magneride accelerometers	Accelerometer	16 Nm	B

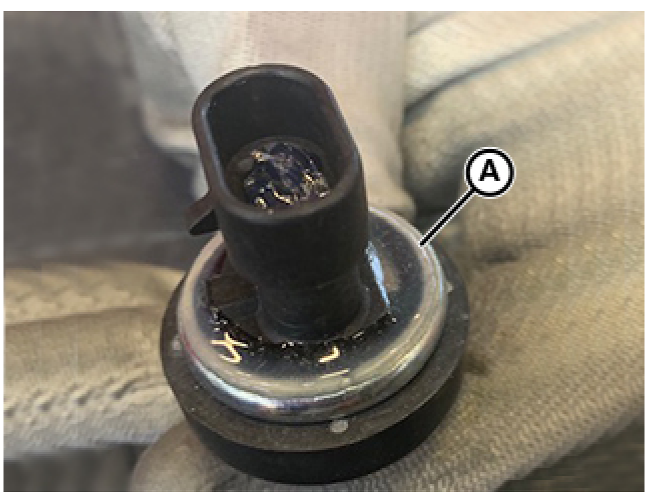
- Disconnect the battery (F2.01).

RH front accelerometer

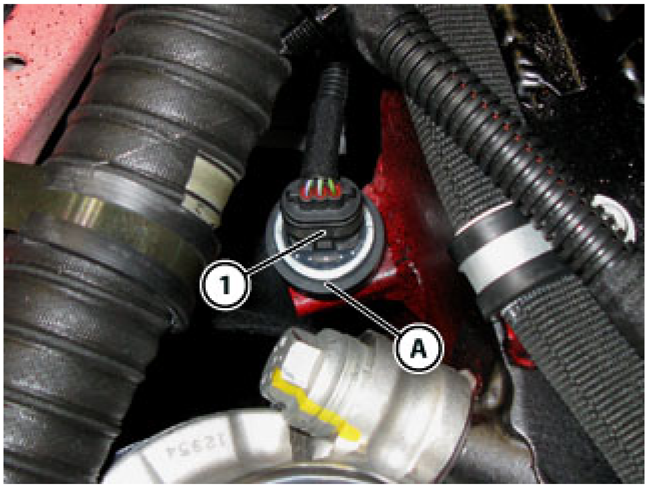
- Remove the engine compartment cosmetic shields (E3.14).
- Remove the right hand lateral cosmetic shield.



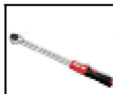
- Disconnect the connector **(1)**.
- Unscrew the accelerometer **(A)** and replace.



- Take the new accelerometer **(A)**.
- Apply enough **NYOGEL 760G - Protective grease for electrical contacts** to cover the pins of the accelerometer **(A)**.



- Screw in the new accelerometer **(A)**.



Tightening torque	Nm	Class
Accelerometer	16 Nm	B

- Connect the connector **(1)**.

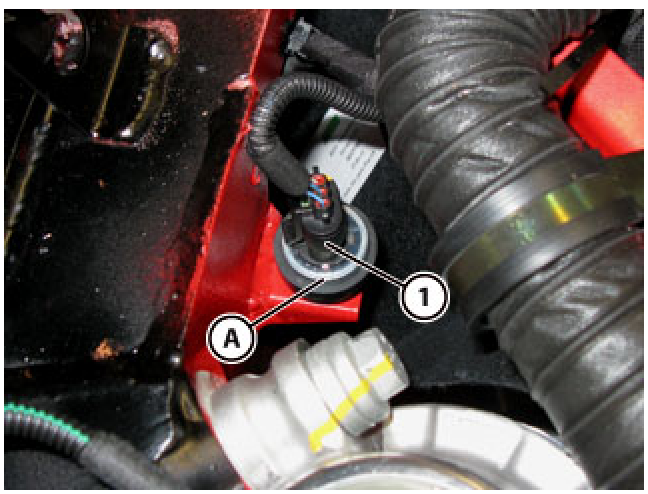
- Refit the engine compartment cosmetic shields (E3.14).

Refit the right hand lateral cosmetic shield.

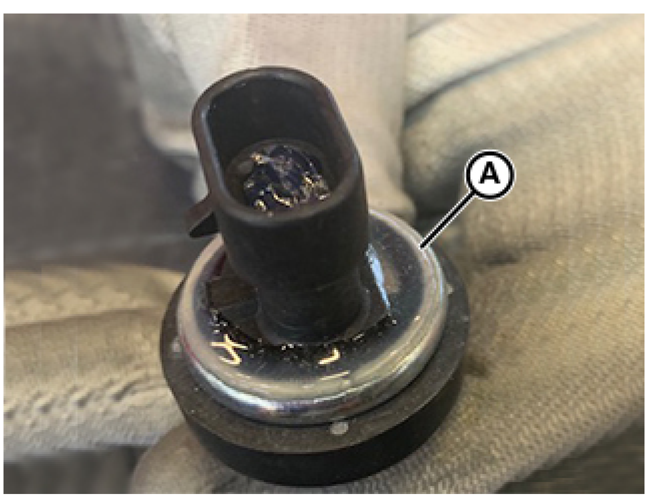
LH front accelerometer

- Remove the engine compartment cosmetic shields (E3.14).

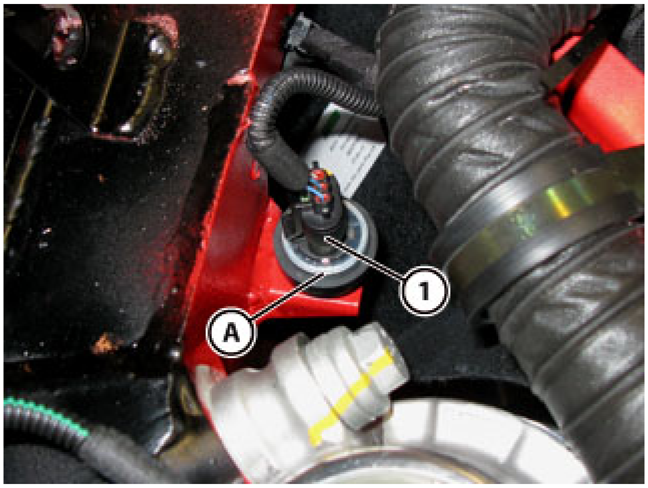
Remove the left hand lateral cosmetic shield.



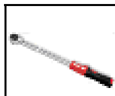
- Disconnect the connector **(1)**.
- Unscrew the accelerometer **(A)** and replace.



- Take the new accelerometer **(A)**.
- Apply enough **NYOGEL 760G - Protective grease for electrical contacts** to cover the pins of the accelerometer **(A)**.



- Screw in the new accelerometer **(A)**.



Tightening torque	Nm	Class
Accelerometer	16 Nm	B

- Connect the connector **(1)**.

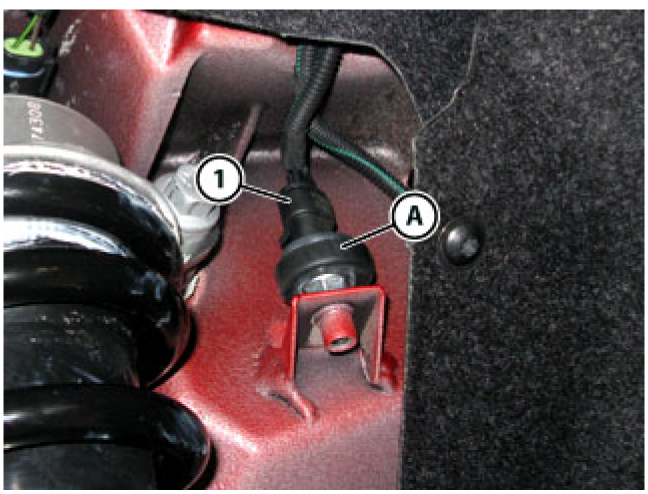
- Refit the engine compartment cosmetic shields (E3.14).

Refit the left hand lateral cosmetic shield.

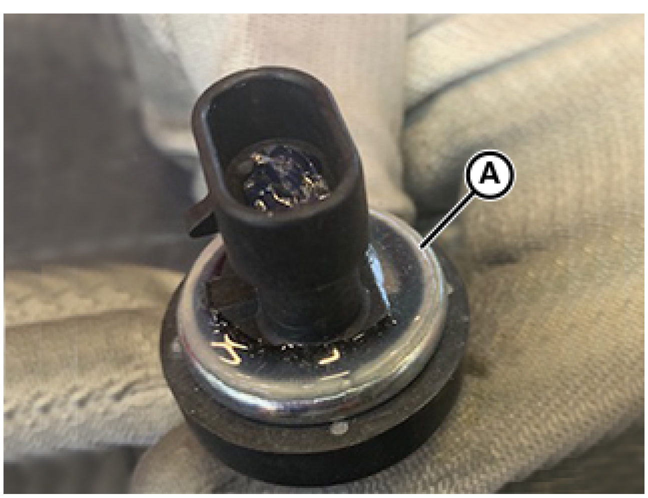
Rear accelerometer

- Remove the complete wheels (D2.01).

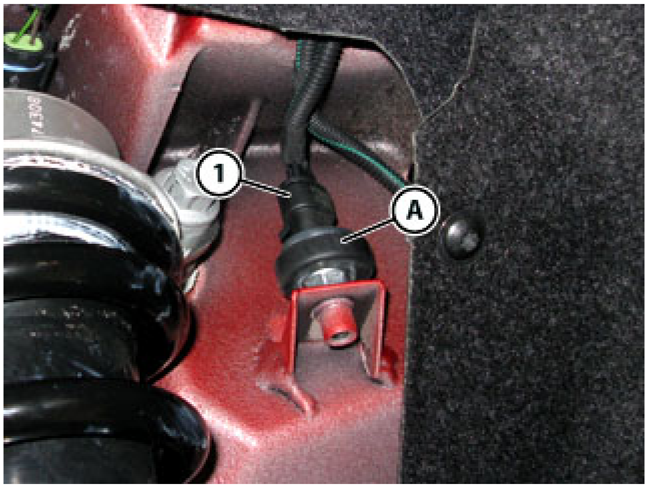
Remove the rear left hand wheel.



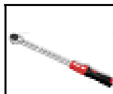
- Disconnect the connector **(1)**.
- Unscrew the accelerometer **(A)** and replace.



- Take the new accelerometer **(A)**.
- Apply enough **NYOGEL 760G - Protective grease for electrical contacts** to cover the pins of the accelerometer **(A)**.



- Screw in the new accelerometer **(A)**.



Tightening torque	Nm	Class
Accelerometer	16 Nm	B

- Connect the connector **(1)**.

- Refit the complete wheels (D2.01).

Refit the rear left hand wheel.

- Connect the battery (F2.01).
- Connect the DEIS tester to the diagnostic socket (F2.09).

  Perform the respective cycle with the DEIS diagnostic tester.